

Carleman–QSVT Method for Solving the Burgers Equation: Method and Results Summary

Based on Burgers_Carlemann_qiskit.ipynb

1 Core idea

The computational pipeline implemented in the notebook is

Burgers PDE $\longrightarrow$ finite difference $\longrightarrow$ Carleman linearization
 $\longrightarrow$ implicit Euler $\longrightarrow$ linear system $\longrightarrow$ QSVT inverse.

The key point is that **QSVT does not directly simulate the original nonlinear Burgers equation**. Instead, the nonlinear problem is first lifted into a truncated linear Carleman system, and QSVT is then used at each time step to approximate the inverse of the resulting linear matrix.

2 Original problem and spatial discretization

The notebook considers the viscous Burgers equation

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u, \quad \nu = 0.01, \quad (1)$$

with initial condition

$$u(0, x) = \sin(2\pi x), \quad x \in [0, 1],$$

and homogeneous boundary conditions. The numerical parameters are

$$N = 5, \quad \Delta t = 0.1, \quad T = 0.3,$$

so the simulation contains three nontrivial time steps.

Central finite differences convert the PDE into a quadratic nonlinear ODE system,

$$\dot{\mathbf{u}} = F_0 + F_1 \mathbf{u} + F_2(\mathbf{u} \otimes \mathbf{u}). \quad (2)$$

Here, F_1 mainly contains the diffusion contribution, while F_2 represents the quadratic nonlinearity of the Burgers convection term.

3 Carleman linearization

Carleman linearization promotes nonlinear monomials to new variables, for example

$$\mathbf{y}_1 = \mathbf{u}, \quad \mathbf{y}_2 = \mathbf{u}^{\otimes 2}, \quad \mathbf{y}_3 = \mathbf{u}^{\otimes 3}, \dots$$

At infinite order, this converts the nonlinear system into an infinite-dimensional linear system. The notebook truncates the expansion at

$$N_T = 2,$$

so that only

$$\mathbf{y} = \begin{pmatrix} \mathbf{u} \\ \mathbf{u} \otimes \mathbf{u} \end{pmatrix}$$

is retained. For $N = 5$, the Carleman dimension is

$$D = N + N^2 = 5 + 25 = 30,$$

consistent with the 30×30 matrix reported by the notebook. For the quantum representation, this is padded to $32 = 2^5$.

Meaning of the truncation: the derivative $d(\mathbf{u}^{\otimes 2})/dt$ generally generates third-order terms $\mathbf{u}^{\otimes 3}$. Therefore, truncation at $N_T = 2$ is already an approximation. The resulting Carleman truncation error is distinct from the QSVT polynomial approximation error.

4 Implicit Euler: converting time evolution into a linear solve

The Carleman system can be written as

$$\dot{\mathbf{y}} = A\mathbf{y} + \mathbf{b}.$$

Using implicit Euler,

$$\frac{\mathbf{y}_{n+1} - \mathbf{y}_n}{\Delta t} = A\mathbf{y}_{n+1} + \mathbf{b},$$

which gives

$$\boxed{(I - \Delta t A)\mathbf{y}_{n+1} = \mathbf{y}_n + \Delta t \mathbf{b}.} \tag{3}$$

Defining

$$M = I - \Delta t A, \quad \mathbf{r}_n = \mathbf{y}_n + \Delta t \mathbf{b},$$

each time step becomes

$$\mathbf{y}_{n+1} = M^{-1}\mathbf{r}_n.$$

Thus, the central quantum task is to approximate M^{-1} .

5 QSVT linear solver

The notebook uses QSVT to apply a polynomial transformation to the singular values of a block-encoded matrix M . The polynomial is designed such that

$$P(x) \approx \frac{1}{x}$$

over the relevant spectral interval, so that ideally

$$P^{(\text{SV})}(M) \approx M^{-1}.$$

The notebook uses

$$\kappa = 8, \quad \text{tolerance} = 10^{-5},$$

and generates

$$\boxed{560 \text{ QSVT phase angles.}}$$

The effective condition numbers of both block-encoding constructions are smaller than 8, so this choice of κ covers the target spectra in both cases.

6 Two block-encoding constructions

The notebook compares two approaches:

1. **Dense block encoding:** the full matrix is embedded directly into a larger unitary;
2. **Pauli-LCU block encoding:** the matrix is expanded as $M = \sum_j \alpha_j P_j$ and implemented through a linear combination of unitaries.

Although the matrix has only a small number of nonzero entries per row in the computational basis, the notebook explicitly observes that its *Pauli decomposition is not sparse*. As a result, a naive Pauli-LCU implementation becomes extremely costly.

7 Resource-estimation results

The notebook reports the following full-circuit estimates:

Method	qubits	α	κ_{eff}	gates	CX	depth
Dense	6	2.240	2.57	2,614,444	996,697	1,971,035
Pauli-LCU	14	4.053	4.65	1,283,229,363	600,388,684	948,235,555

The Pauli-LCU total gate count is approximately

$$\frac{1.283 \times 10^9}{2.614 \times 10^6} \approx 4.9 \times 10^2,$$

i.e. about **500 times larger** than the dense implementation. This illustrates that

matrix sparsity $\nrightarrow$ Pauli-basis sparsity.

For this example, block encoding itself is already a major resource bottleneck.

8 Interpretation of the numerical results

The notebook compares three trajectories: a classical implicit solver, Dense-QSVT, and Pauli-LCU-QSVT.

Burgers profile

The initial sine profile is progressively steepened by the nonlinear convection term $u u_x$, while the diffusion term νu_{xx} simultaneously smooths the profile. The space-time plots therefore show the competition between nonlinear steepening and viscous smoothing.

RMSE

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_j \left(u_j^Q - u_j^{\text{classical}} \right)^2}$$

measures the **absolute numerical error** between the QSVT result and the classical reference solution.

Amplitude-encoding fidelity

The notebook uses

$$F = \left| \frac{\mathbf{u}_{\text{cl}}^T \mathbf{u}_Q}{\|\mathbf{u}_{\text{cl}}\| \|\mathbf{u}_Q\|} \right|^2.$$

This measures whether the normalized amplitude vectors point in the same direction, or equivalently whether their shapes agree after normalization. Thus, even if $\mathbf{u}_Q = c \mathbf{u}_{\text{cl}}$, one can still have $F = 1$. Fidelity and RMSE therefore characterize different aspects of the error.

Success probability

Block encoding and QSVT generally require post-selection on ancilla qubits. The success probability measures the probability of obtaining the component proportional to $M^{-1}|b\rangle$. If this probability is small, the algorithm requires more repetitions or amplitude amplification. Therefore, high fidelity alone does not imply a low practical quantum cost.

9 What does the notebook actually demonstrate?

The most appropriate conclusions are:

1. It establishes a complete computational pipeline

nonlinear PDE $\rightarrow$ Carleman linear system $\rightarrow$ QSVT linear solve.

2. Dense-QSVT and Pauli-LCU-QSVT are both compared against the same classical implicit reference, providing a consistency check of the quantum linear-solving step.
3. The resource estimates show that **“QSVT can formally solve the linear system” does not imply that the full quantum algorithm is efficient**. Even for a small $D = 30$ problem, the full circuit already reaches gate counts of order 10^6 – 10^9 .
4. The main bottlenecks are not limited to matrix inversion; they also include Carleman dimension growth, block-encoding normalization and complexity, and the QSVT polynomial degree.

10 Main limitations

- **Low Carleman order:** $N_T = 2$ discards third-order and higher monomials, which may become important for stronger nonlinearities or longer evolution times.
- **Very low spatial resolution:** $N = 5$ is suitable for a proof of principle but is not sufficient for a high-accuracy benchmark of shock-like Burgers dynamics.
- **Carleman dimension growth:**

$$D = N + N^2 + \dots + N^{N_T},$$

which grows rapidly with the truncation order.

- **Large circuit cost:** even for the present 30×30 linear system, the QSVT and block-encoding circuits are extremely deep. The notebook should therefore be viewed as an algorithmic demonstration rather than a demonstration of near-term quantum advantage.

One-sentence summary

The notebook demonstrates a complete route from a nonlinear Burgers PDE, through finite-difference discretization and truncated Carleman linearization, to a sequence of linear systems solved by QSVT; the numerical pipeline is self-consistent, but the resource estimates reveal that Carleman dimension growth, block encoding, and QSVT circuit depth are the key obstacles that must be addressed before this approach can offer practical quantum advantage.